Week 5: Comprehensive Project Report & Strategic Recommendations

NIFTY 50 Index: From Raw Data to Predictive Modelling (June 2016 to June 2026)

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Dataset: NIFTY 50 index daily prices, 1 June 2016 to 30 June 2026 (2,483 trading days)

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, Scikit-learn), Jupyter

Executive Summary

This project took ten years of NIFTY 50 daily data (2,483 trading days, June 2016 to June 2026) through the full data science cycle: acquisition and cleaning, visual storytelling, statistical testing and predictive modelling. Four findings stand out.

- Strong but uneven growth. The index rose about 192% (roughly 11% a year, excluding dividends), but it fell 38.4% from peak to trough in early 2020 and regained its high in about seven and a half months. A smaller decline of up to 15% occurred in the first half of 2026.
- A small, real positive drift. The average day earned +0.048% (95% confidence interval 0.008% to 0.089%), statistically different from zero but tiny relative to daily noise (standard deviation 1.02%).
- Risk comes in bursts. Volatility in 2020 was about 2.6 times that of 2017 to 2019 ($p < 0.001$), and extreme days are far more common than a normal distribution implies.
- The next-day direction could not be predicted. Two models scored 49.2% and 48.4% against a 50.3% baseline, and no robust weekday effect was found. This is a plausible outcome for an efficient market, not a failure of the method.

The practical message is that this data supports long-horizon, risk-aware investing and risk management, and gives no support for short-term timing based on simple past-price signals. This report is an academic exercise and does not constitute investment advice.

1. Introduction and Scope

The goal was to show the end-to-end workflow on one coherent dataset and to turn technical results into conclusions that decision-makers can use. The NIFTY 50 was chosen for its relevance to Indian markets, its public availability and its mix of calm and extreme periods. The analysis is at the index level and uses daily Open, High, Low, Close, Adjusted Close and Volume from Yahoo Finance.

2. Methodology Summary

Week	Focus	Main methods	Key output
1	Acquisition, cleaning, EDA	Quality audit, type fixes, recoding zero volume, feature engineering	Cleaned dataset; 4 charts
2	Visual storytelling	Trend, rolling volatility, annual returns, drawdown, heatmap, distribution	6-chart narrative
3	Hypothesis testing	t-test, Levene, Welch, ANOVA, Tukey, Pearson, confidence interval	4 hypotheses tested
4	Machine learning	Logistic regression, decision tree, chronological split	Next-day direction models

3. Results by Stage

3.1 Data Preparation (Week 1)

The raw data was in good condition: no missing values, no duplicates and no inconsistent prices. The issues found were dates stored as text and 28 days (1.1%) with a volume of zero, which were recoded as missing because a real index volume of zero is implausible. No rows were dropped, so the time series stayed continuous.

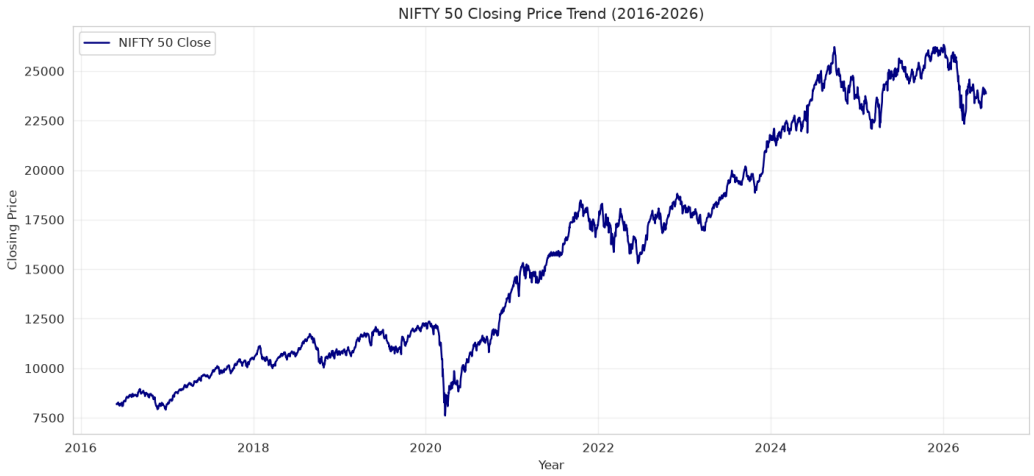


Figure 1: NIFTY 50 closing price, 2016 to 2026

The index moved from 8,180 to 23,866. Daily returns average +0.048% with a standard deviation of 1.02%, a skewness of -1.07 and an excess kurtosis of 19.2, so tails are heavy and tilted toward falls.

3.2 Visual Storytelling (Week 2)

Six charts told the story from growth to risk. Annual returns were positive in ten of eleven periods (the eleventh being the part-year 2026), and even 2020 ended up 14.8% despite the crash. Volatility peaked at 4.74% (30-day standard deviation) on 23 April 2020.

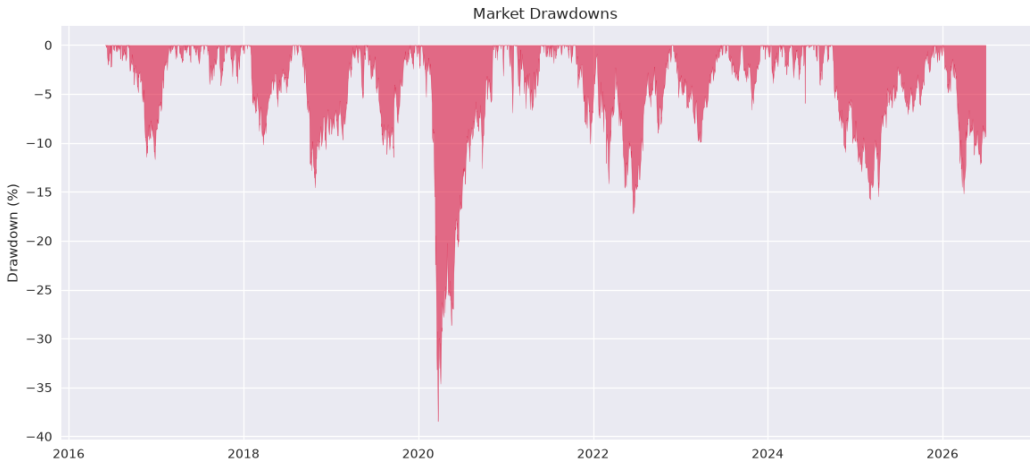


Figure 2: Drawdown from the previous high. The 2020 fall (-38.4%) is the deepest; 2026 reached -15.2%

Most other years contained declines of 10% to 17%, so a double-digit fall from a high is normal, while a fall near 40% is rare in this decade.

3.3 Statistical Testing (Week 3)

Hypothesis	Result	Conclusion
H1: mean daily return differs from 0	t = 2.36, p = 0.018	Yes: +0.048% a day, 95% CI 0.008% to 0.089%
H2: 2020 more volatile than 2017-2019	Levene p = 8e-21; Welch p = 3e-10	Yes: spread about 2.6 times larger
H3: weekday effect	ANOVA p = 0.049; Kruskal p = 0.39	Not robust: no significant pair in post-hoc test
H4: volume vs size of moves	r = 0.37, p = 8e-81	Yes, moderate: volume explains about 14%

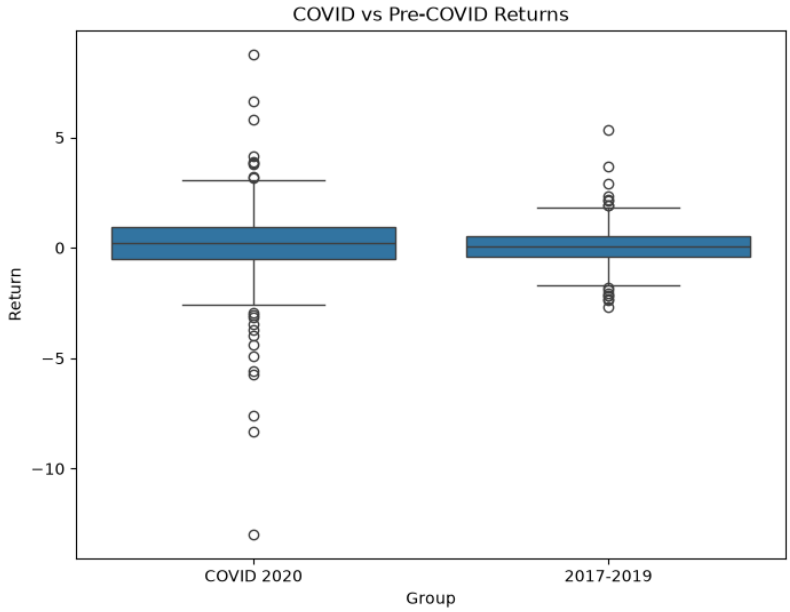


Figure 3: Daily returns in 2020 versus 2017 to 2019

3.4 Predictive Modelling (Week 4)

A logistic regression and a depth-3 decision tree predicted next-day direction from lagged returns, price-to-moving-average ratios and recent volatility, using a chronological 80/20 split (457 test days).

Model	Test accuracy	Note
Majority-class baseline	50.3%	Test set almost balanced
Logistic regression	49.2%	Predicted Up on 447 of 457 days; AUC about 0.49
Decision tree (depth 3)	48.4%	Balanced predictions, no skill

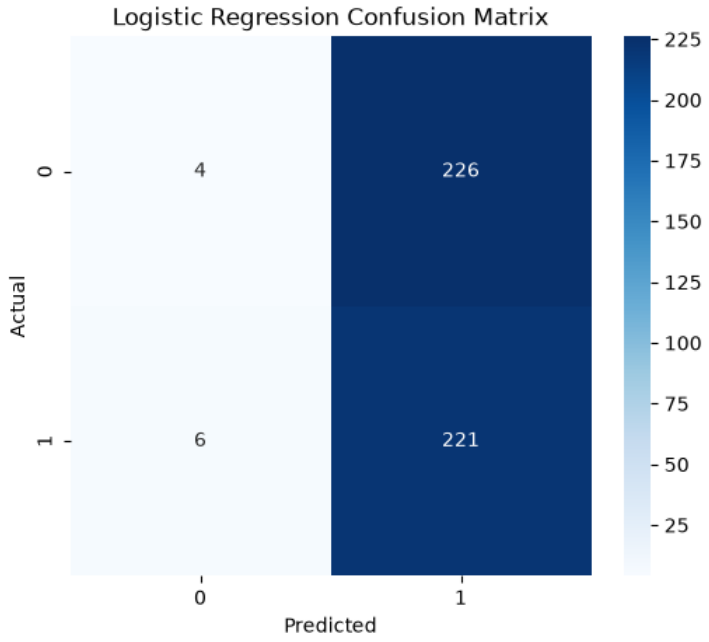


Figure 4: Logistic regression confusion matrix (0 = Down, 1 = Up)

4. Integrated Insights

- Growth and risk are inseparable. The decade's 192% gain contains the one event that shaped the whole story, the 2020 crash. Seen from far away the path is smooth, and up close it is violent.
- Different findings operate at different time scales. A tiny positive daily drift (H1) compounds into large long-run gains, while the very same noise makes a single day almost impossible to forecast (Week 4). These results are consistent, not contradictory.
- Cleaning decisions had downstream effects. Recoding zero volumes in Week 1 is what allowed the volume test in Week 3 to be run on valid data, and expressing moving averages as ratios in Week 4 removed a trend-related distortion in the model inputs.
- Negative results are results. The absence of a robust weekday effect and of next-day predictability rules out tempting but unsupported trading ideas.

5. Strategic Recommendations

5.1 For Long-Term and Retail Investors

- Favour a long holding period and regular investing over short-term timing. The positive drift is real but small per day and only emerges over long horizons.
- Plan for drawdowns before they happen. A fall of about 38% occurred once in this decade and falls of 10% to 17% occurred in most years, so position sizes and emergency savings should allow holding through them without forced selling.
- Treat calendar rules (such as "avoid Mondays") sceptically; the evidence for them here is not robust.

5.2 For Risk Managers and Portfolio Teams

- Use risk measures that adapt over time (for example rolling or GARCH-type volatility) because risk clusters in bursts; a single average volatility understates risk in turbulent months.
- Calibrate stress tests and tail hedges to the observed extreme days (a skewness of -1.07 and an excess kurtosis of 19.2) instead of a normal distribution.
- Use volume as an early-warning indicator: it is moderately linked to large price moves ($r = 0.37$), although not enough to forecast them.

5.3 For Product, Advisory and Analytics Teams

- Show clients drawdown charts alongside annual returns. In this dataset 2020 finished up 14.8% while containing a 38% fall, and annual figures alone hide that.
- Do not build a short-term direction-prediction feature on price history alone. The evidence says it would not outperform a coin toss; place effort where information is genuinely new, such as news and sentiment.

5.4 For the Data Science Team (Next Steps)

- Re-run Week 4 on a 10-day or 20-day horizon, with external features (India VIX, US overnight returns, news sentiment) and walk-forward validation.
- Connect this work to the market-news classification project, testing whether news categories explain the large-move days found here.
- Extend the analysis from the index to constituent stocks to find which names drove the 2020 and 2026 declines.

6. Potential Business Impact

For an investment-advisory or wealth platform, the findings point to three concrete improvements: risk communication built on drawdowns and not only returns, volatility-aware portfolio limits instead of fixed ones, and a decision not to invest in short-term price-only prediction products. For a trading or analytics team, the null result saves effort and sets a clear bar: any future signal must beat the 50.3% baseline on a chronological test set before it deserves consideration.

7. Limitations and Future Work

- One index and one decade, with a single major crash; conclusions about crash behaviour rest on one event, and the 2016 and 2026 annual figures are part-years.
- Yahoo's index volume is not the official NSE volume (its minimum value is 100 shares), so volume results are indicative.
- Statistical tests assume more independence than daily returns have because volatility clusters; p-values are approximate. The weekday result in particular is borderline and sensitive to 2020.
- The 2020 comparison group (2017 to 2019) and the crisis year were chosen from prior knowledge, not from a formal change-point method.
- The models used six price-derived features and one split; a 457-day test set cannot separate small differences in accuracy.
- Prices are price-index values and exclude dividends, so total returns would be somewhat higher.

8. Conclusion

The NIFTY 50 rewarded patient investors over this decade, but not smoothly and not predictably in the short run. The data confirm a small positive drift, strongly clustered volatility and heavy tails, and they offer no support for simple weekday or next-day timing rules. Taken together, the five weeks show a complete workflow, from a carefully audited dataset to conclusions that are stated with the strength the evidence allows.